



PAT: Accelerating LLM Decoding via Prefix-Aware Attention with Resource Efficient Multi-Tile Kernel

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<https://github.com/flashserve/PAT>



<https://www.arxiv.org/abs/2511.22333>

Background and Introduction

LLM serving is dominated by memory-bound decode attention. Existing attention implementations (e.g., FlashAttention) fail to exploit substantial and hierarchical shared prefixes during decode attention execution, leading to massive redundant memory access from global memory to on-chip memory.

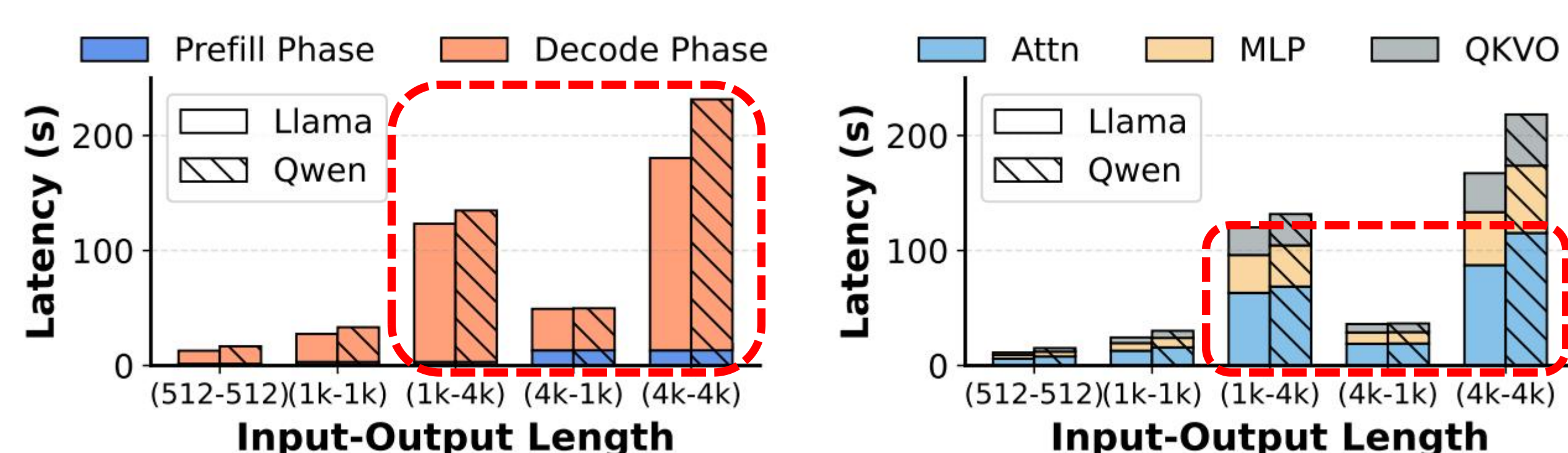


Fig 1. Latency breakdown of Llama-3-8B and Qwen3-8B.

PAT (1) aggregates queries with shared KV blocks to reduce redundant global memory access and (2) executes attention with multi-tile kernels to improve hardware utilization.

PAT reduces kernel latency by 54% and TPOT by 17-93%.

Background: Memory Bottleneck of Decode Attention

Decode attention is memory-bound due to heavy KV cache loads from slow global memory to on-chip memory.

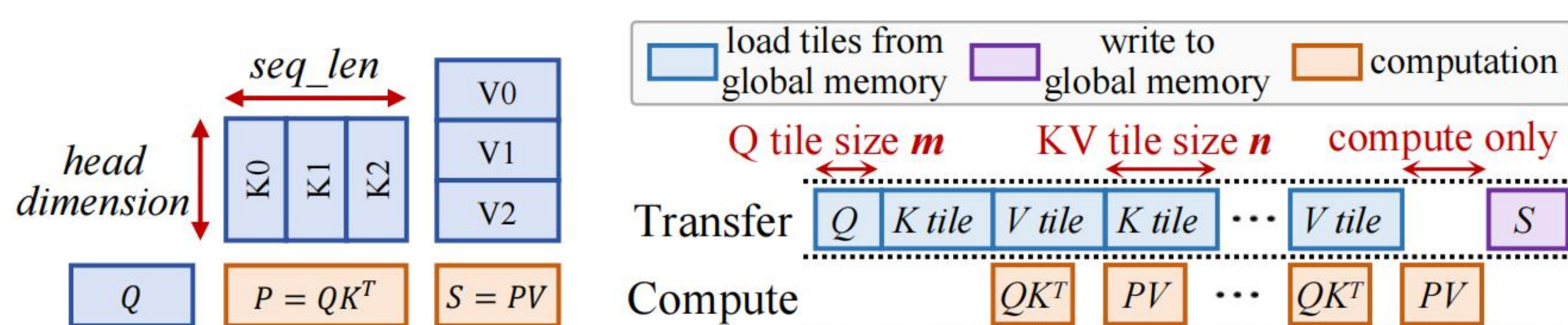


Fig 2. Tiling execution pipeline of decode attention

Motivation: Shared Prefixes and Redundant Mem Access

Shared prefixes are common and hierarchical in LLM serving. Under continuous batching, cross-request prefix sharing brings multi-level intra-batch prefixes, which covers 2.8-83% of KV caches under real-world traces.

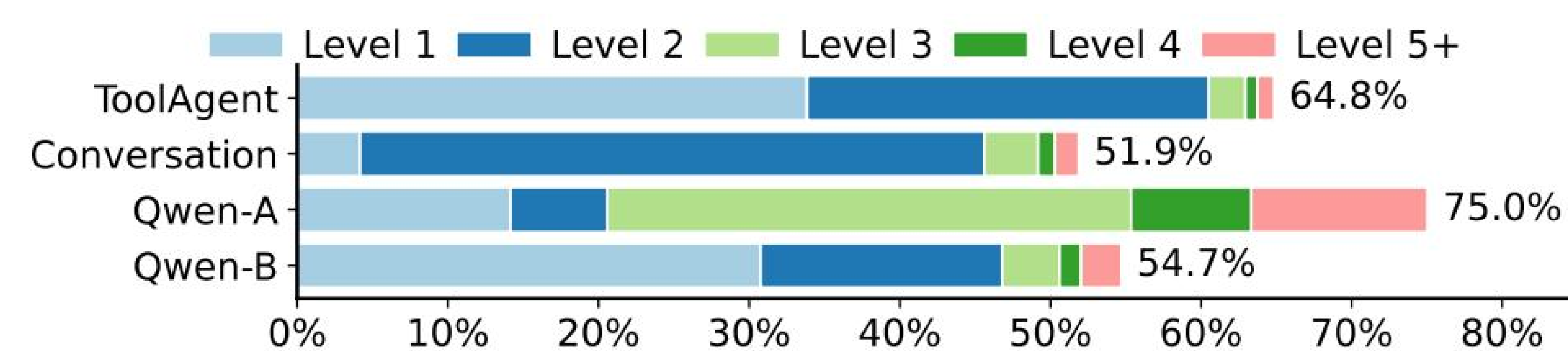


Fig 3. Prefix ratio of Mooncake traces and Qwen traces.

Existing attention implementations fail to exploit intra-batch prefix sharing and stall memory-bound decode attention.

- Inefficiency 1: Redundant Memory Access
- Inefficiency 2: Resource Underutilization

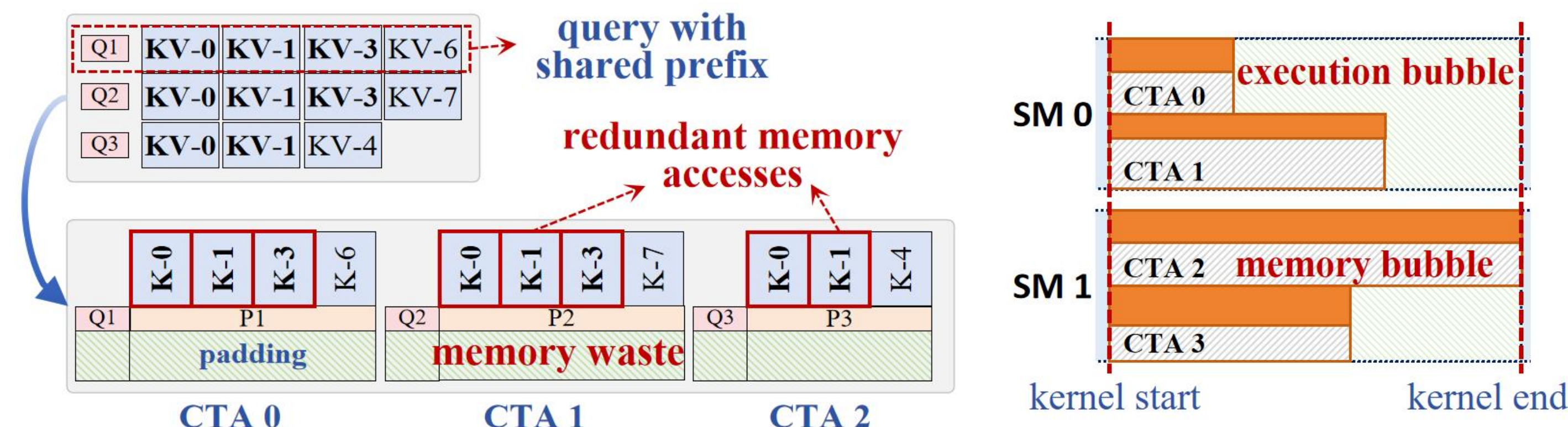


Fig 4. Existing query-centric packing strategy for decode batch.

Attention Kernel Design

Insight: Aggregating prefixes avoids redundant memory access

1. Pack: Group queries by shared prefix into one CTA
2. Forward: Execute with resource-efficient Multi-Tile Kernel
3. Merge: Combine results via online softmax

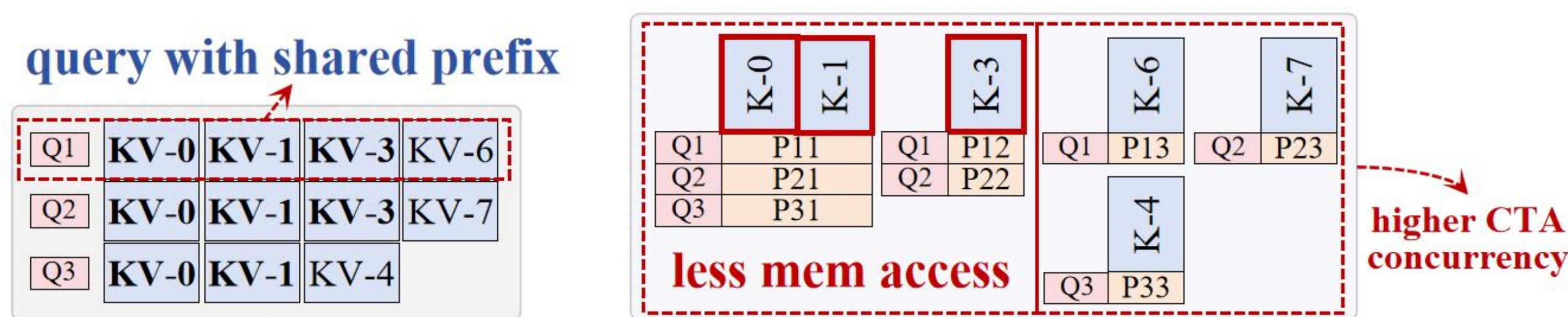


Fig 5. PAT's memory-centric packing strategy.

Design (Pack): Prefix-Aware Packing

PAT packs queries into shared-prefix CTAs to reduce memory loads via memory-centric pack scheduler. It overlaps scheduling overhead with pre-attention tasks (metadata prepare, QKV proj.).

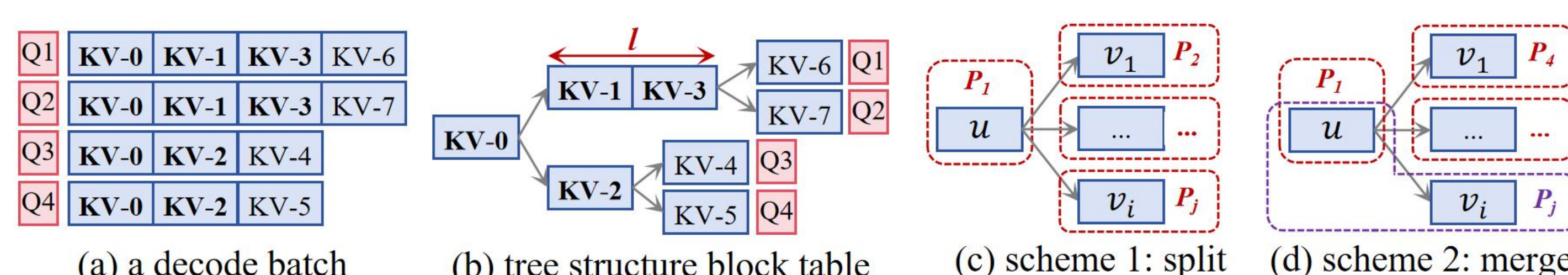


Fig 6. Workflow of the pack scheduler: (a) input decode batch; (b) build tree structure block table; (c) and (d) various packing schemes.

Design (Forward): Multi-tile Kernel

Offline: PAT automatically derives equivalent tile size with high bandwidth and low kernel latency.

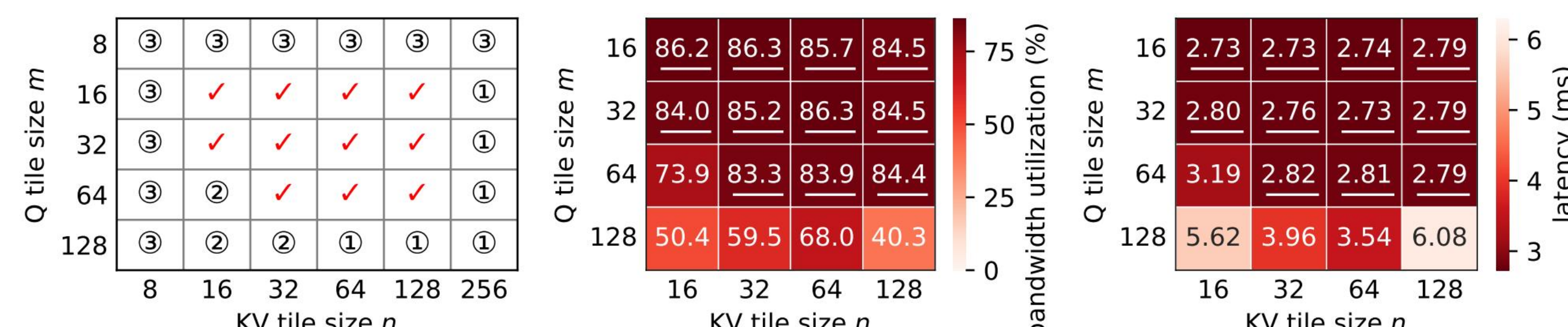


Fig 7. Equivalent tile sizes on A100 and their verification.

Online: PAT selects proper tile size for each CTA (pack) to reduce execution bubble both on time and space dimensions.

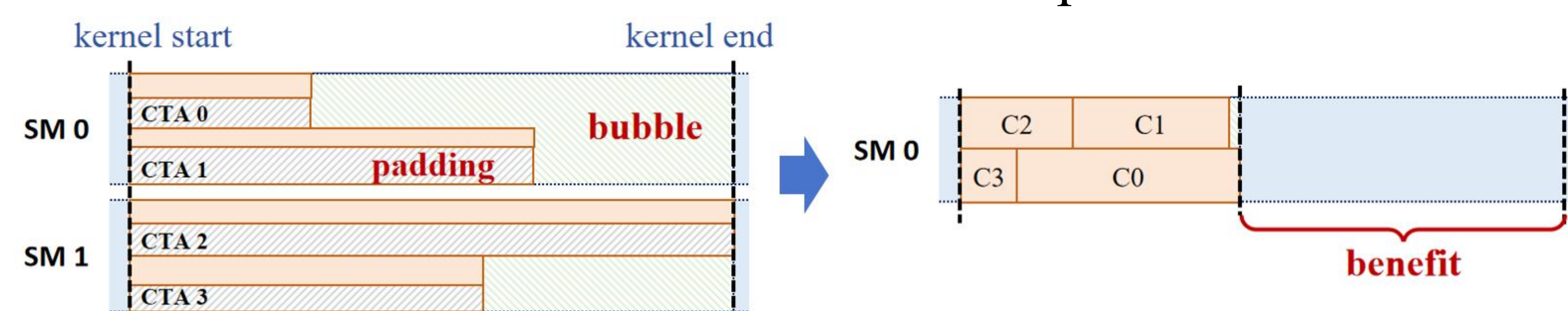


Fig 8. Mitigate execution bubble with multi-tile kernel.

Design (Merge): Output Merge

PAT merges outputs of each query with online softmax kernel.

Evaluation on A100 and H100

PAT: an off-the-shelf plugin for vLLM

Evaluation on both real-world and synthetic workloads shows that PAT reduces attention latency by 53.5% on average (left) and TPOT by 17.0-93.1% under the same configurations (right) against state-of-the-art attention kernels.

